

We need the last two digits of

$$9^{9^9} \bmod 100$$

Step 1: Use Euler's Totient Function

$$\varphi(100) = 40$$

Since $\gcd(9, 100) = 1$, Euler's theorem gives:

$$9^{40} \equiv 1 \pmod{100}$$

So we reduce the exponent:

$$9^{9^9} \equiv 9^{(9^9 \bmod 40)} \pmod{100}$$

Step 2: Compute $9^9 \bmod 40$

Note:

$$9^2 = 81 \equiv 1 \pmod{40}$$

So:

$$9^9 = (9^2)^4 \cdot 9 \equiv 1^4 \cdot 9 \equiv 9 \pmod{40}$$

Step 3: Final reduction

$$9^{9^9} \equiv 9^9 \pmod{100}$$

Now compute:

$$\begin{aligned} 9^2 &= 81 \equiv -19 \pmod{100} \\ 9^4 &= 81^2 = 6561 \equiv 61 \pmod{100} \\ 9^8 &= 61^2 = 3721 \equiv 21 \pmod{100} \\ 9^9 &= 21 \cdot 9 = 189 \equiv 89 \pmod{100} \end{aligned}$$

Final Answer:

$$\boxed{89}$$

So, the last two digits of 9^{9^9} are 89.

How to evaluate

$$\phi(100)$$

We use the Euler's Totient Function, which counts numbers $\leq n$ that are coprime to n .

Step 1: Prime factorization of 100

$$100 = 2^2 \times 5^2$$

Step 2: Formula for totient

For:

$$\begin{aligned} n &= p^a \cdot q^b \\ \phi(n) &= n \left(1 - \frac{1}{p}\right) \left(1 - \frac{1}{q}\right) \end{aligned}$$

Step 3: Apply to 100

$$\begin{aligned} \phi(100) &= 100 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) \\ &= 100 \cdot \frac{1}{2} \cdot \frac{4}{5} \\ &= 100 \cdot \frac{4}{10} = 40 \end{aligned}$$

Ques-2:- Given a is an odd integer.

To prove $a^2 \equiv 1 \pmod{8}$, we will use Mathematical induction. [Here, $a = 2n+1$]

for $a = 1$ i.e. $n = 0$.

$$1^2 \equiv 1 \pmod{8}$$

which is true.

Now, let us assume that for $a = 2k+1$, it is true. i.e. $(2k+1)^2 \equiv 1 \pmod{8}$ ——— ①

Now, we will prove that above will be true for $n = k+1$ also.

$$\text{i.e. } (2k+1)^2 \equiv 1 \pmod{8}$$

$$\left[\because (2k+1)^2 = (2k-1)^2 + 8k \right]$$

$$\Rightarrow (2k-1)^2 + 8k \equiv 1 \pmod{8}$$
$$1 + 0 \equiv 1 \pmod{8}$$

[Using eq. ①]

Hence, $\boxed{a^2 \equiv 1 \pmod{8}}$

Ques-3:- (a) $9 + 8 \pmod{10}$
 $= 17 \pmod{10}$
 $= 7 \pmod{10}$

(b) $85 - 28 \pmod{20} = 57 \pmod{20} = 17 \pmod{20}$

OR $= [85 \pmod{20} - 28 \pmod{20}] \pmod{20}$
 $= 5 - 8 \pmod{20}$
 $= -3 \pmod{20} \text{ or } 17 \pmod{20}$

$$\begin{aligned}
 (c) \quad & 91 \times 89 \pmod{30} \\
 &= [91 \pmod{30} \times 89 \pmod{30}] \pmod{30} \\
 &= (1 \times -1) \pmod{30} \\
 &= -1 \pmod{30} \\
 &= 29 \pmod{30}
 \end{aligned}$$

Ques-4:- (a) $\because a \equiv b \pmod{n}$
 $\Rightarrow a - b = k \cdot n \quad \text{--- (1)}$

and $c \equiv c \pmod{n}$

$$\Rightarrow c - c = 0 \cdot n \quad \text{--- (2)}$$

Adding (1) & (2),

$$(a - b) + (c - c) = k \cdot n + 0 \cdot n$$

or $(a + c) - (b + c) = k \cdot n$

or $a + c \equiv b + c \pmod{n}$

Multiplying ~~by~~ a & b ,

$$\begin{aligned}
 ac &= (kn + b)(0 \cdot n + c) \\
 &= kc \cdot n + bc
 \end{aligned}$$

$$ac - bc = kc \cdot n$$

$$ac - bc = k_1 \cdot n$$

[where $kc = k_1$]
 $\downarrow$
 an integer.

$$ac \equiv bc \pmod{n}$$

(b) To prove this part we use induction argument.

i.e. for $k=1$,

$$a \equiv b \pmod{n}$$

which is true.

We will assume it is true for some fixed k .

And, $\therefore a \equiv b \pmod{n}$

$$a^k \equiv b^k \pmod{n}$$

$$\Rightarrow aa^k \equiv bb^k \pmod{n} \quad \left[\begin{array}{l} \text{Using the prop.} \\ \text{if } a \equiv b \pmod{n} \\ \& c \equiv d \pmod{n} \\ \text{then } ac \equiv bd \pmod{n} \end{array} \right]$$

$$\Rightarrow a^{k+1} \equiv b^{k+1} \pmod{n}$$

Hence, the induction steps proves that

$$a^k \equiv b^k \pmod{n}$$

Ques-5 :-

(a)

$$1475 = 1 \times 1200 + 275$$

$$1200 = 4 \times 275 + 100$$

$$275 = 2 \times 100 + 75$$

$$100 = 1 \times 75 + \boxed{25}$$

$$75 = 3 \times 25 + 0$$

Hence 25 is the GCD of 1475 and 1200.

OR

q	r	s	t
1	1475	1200	275
4	1200	275	100
2	275	100	75
1	100	75	25
3	75	25	0
	$\boxed{25}$	0	

(b)

$$766, 1235$$

$$1235 = 1 \times 766 + 469$$

$$766 = 1 \times 469 + 297$$

$$469 = 1 \times 297 + 172$$

$$297 = 1 \times 172 + 125$$

$$172 = 1 \times 125 + 47$$

$$125 = 2 \times 47 + 31$$

$$47 = 1 \times 31 + 16$$

$$31 = 1 \times 16 + 15$$

$$16 = 1 \times 15 + 1$$

$$15 = 15 \times 1 + 0$$

Hence GCD of 766 and 1235 is 1.

Ques-6 :-

(a)

$$272 = 2 \times 119 + 34$$

$$119 = 3 \times 34 + 17$$

$$34 = 2 \times 17 + 0$$

_____ (1)

_____ (2)

_____ (3)

$$\text{GCD}(272, 119) = 17$$

$$17 = 119x + 272y$$

To find x & y ,

$$17 = 119 - 3 \times 34$$

[from (2)]

$$17 = 119 - 3 \times (272 - 2 \times 119) \quad [\text{from (1)}]$$

$$17 = 7 \times 119 - 3 \times 272$$

$$\text{or } \boxed{17 = 119(7) + 272(-3)}$$

Hence $x = 7$ and $y = -3$

(b)

$$\text{GCD}(1769, 2378)$$

$$2378 = 1 \times 1769 + 609 \quad \text{--- (1)}$$

$$1769 = 2 \times 609 + 551 \quad \text{--- (2)}$$

$$609 = 1 \times 551 + 58 \quad \text{--- (3)}$$

$$551 = 9 \times 58 + \boxed{29} \quad \text{--- (4)}$$

$$58 = 2 \times 29 + 0$$

$$\Rightarrow \text{GCD}(1769, 2378) = 29$$

Now find x & y s.t.

$$29 = 1769x + 2378y$$

$$29 = 551 - 9 \times 58 \quad [\text{Using (4)}]$$

$$29 = 551 - 9 \times (609 - 1 \times 551) \quad [\text{Using (3)}]$$

$$29 = 10 \times 551 - 9 \times 609$$

$$29 = 10 \times (1769 - 2 \times 609) - 9 \times 609 \quad [\text{Using (2)}]$$

$$29 = 10 \times 1769 - 29 \times 609$$

$$29 = 10 \times 1769 - 29 (2378 - 1 \times 1769) \quad [\text{Using (1)}]$$

$$\boxed{29 = 1769(39) + 2378(-29)}$$

$$\Rightarrow x = 39 \text{ and } y = -29.$$

Ques-7 :- (a) $18x \equiv 30 \pmod{42}$

Since $\text{GCD}(18, 42) = 6$ and $6 \mid 30$
 $\Rightarrow$ Given linear congruence have 6 solⁿs. and can be written as

$$3x \equiv 5 \pmod{7}$$

$$x \equiv 4 + \left(\frac{42}{6}\right)t \equiv 4 + 7t \pmod{42} \quad [t=0, 1, 2, 3, 4, 5]$$

$$x = 4, 11, 18, 25, 32, 39.$$

(b) $9x \equiv 21 \pmod{30}$

Since $\gcd(9, 30) = 3$ and $3|21$, Hence there must be three incongruent solutions:

$$3x \equiv 7 \pmod{10}$$

$$x \equiv 9 + \left(\frac{30}{3}\right)t \equiv 9 + 10t, \quad t = 0, 1, 2$$

$$x = 9, 19, 29.$$

Ques-8 :- $Z_{15} = \{0, 1, 2, 3, \dots, 14\}$

Multiplicative inverse,

$$ab \equiv 1 \pmod{15} \quad \text{where } a, b \in Z_{15}$$

$$1 \times 1 \equiv 1 \pmod{15}$$

$$2 \times 8 \equiv 1 \pmod{15}$$

$$4 \times 4 \equiv 1 \pmod{15}$$

$$7 \times 13 \equiv 1 \pmod{15}$$

$$11 \times 11 \equiv 1 \pmod{15}$$

$$14 \times 14 \equiv 1 \pmod{15}$$

3, 5, 6, 9, 10, 12 will not have their multiplicative inverse in Z_{15} as these integers are not relatively prime to 15.

Ques-9:-

$$C = P * K \pmod{26}$$

1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L
13	14	15	16	17	18	19	20	21	22	23	24
M	N	O	P	Q	R	S	T	U	V	W	X
				25	26						
				Y	Z						

Now, given $K = 5$

9 14 4 9 1
I N D I A

$$C = P * K \pmod{26}$$

P = I N D I A
9 14 4 9 1

$$P * K = C = 9 \times 5 \quad 14 \times 5 \quad 4 \times 5 \quad 9 \times 5 \quad 1 \times 5 \pmod{26}$$

C = 19 18 20 19 5
S R T S E

$C = \text{SRTSE}$